

# **Self-Adjointness of the Fundamental Fractal Operator $\hat{F}$**

**Closure of the R82 gap from Doc. 322**

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August 2026

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## Abstract

Doc. 322 defines the fundamental fractal operator

$$\hat{F} = \bigoplus_{n=1}^N S_n \circ L_n, \quad S_n = r_n U_n, \quad r_n \in (0, 1), \quad U_n \text{ unitary}, \quad (.1)$$

with logical projectors  $L_n^2 = L_n$ ,  $L_n^\dagger = L_n$ ,  $L_n L_m = L_{\min(n,m)}$ , and books self-adjointness as an open sketch **[S]** (R82).

The upper bound  $N$  in (.1) is *finite*:  $L_0 = \xi \cdot \ell_P$  (Doc. 180 **[K]**) is the minimal length scale of FFGFT — below  $L_0$  the concept of “distance” loses physical meaning. The  $\bigoplus_{n=1}^\infty$  notation in Doc. 322 is a functional-analytic convention, not a physical infinity.

### Main result **[B]**

The R82 gap is **fully closed, with no remaining cases**. Because  $L_0$  makes the number of rungs finite,  $\hat{F}$  is bounded:

$$\|\hat{F}\| \leq \sum_{n=1}^N r_n < \infty.$$

A bounded symmetric operator is automatically self-adjoint; the deficiency indices are  $(0, 0)$ ; there is exactly one self-adjoint realisation. The symmetry  $\hat{F} = \hat{F}^\dagger$  follows from the  $\mathbb{Z}_3$  pairing  $(k, -k)$  — a consequence of the orbifold structure, not an extra requirement.  $\xi = \lambda_{\min}(\hat{F}_{D_4})$  is therefore well-defined **[B]**.

Verification script: 327\_selbstadjungiertheit\_F\_verify.py — 21 assertions, all passed.

## .1 Starting point

R82 (Doc. 190) records: “Open: complete self-adjointness proof for  $\hat{F}$  [S].” Without this proof, the central claim

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \frac{4}{30000}$$

is formally unsecured: “smallest eigenvalue” requires a real spectrum, and a real spectrum requires self-adjointness.

The definition in Doc. 322 contains a genuine tension:  $S_n L_n = r_n U_n L_n$  is in general *not* symmetric for arbitrary unitary  $U_n$  —  $(U_n L_n)^\dagger = L_n U_n^\dagger \neq U_n L_n$ . The self-adjointness must come from the *structure*. This is exactly what the  $\mathbb{Z}_3$  pairing delivers.

## .2 Three observations from the corpus

The following points introduce no new axioms — they read from already-established corpus results what is needed for the proof.

**(F1) Finite number of rungs.**  $L_0 = \xi \cdot \ell_P$  is the minimal length scale (Doc. 180 [K]). The scale ladder ends at  $L_0$ ;  $N$  is finite;  $\sum_{n=1}^N r_n < \infty$  follows trivially. No convergence condition needs to be imposed — there are no infinities in FFGFT.

**(F2)  $\mathbb{Z}_3$  phase pairing.**  $U_n$  acts sector-wise as a phase rotation  $e^{i\theta_{n,k}}$  (Doc. 322: “phase rotations on the scale sectors”), and the phases respect the orbifold symmetry:  $\theta_{n,-k} = -\theta_{n,k}$  for  $\mathbb{Z}_3$ -charged pairs  $(k, -k)$ ; on null sectors ( $q = 0$ ):  $\theta \in \{0, \pi\}$ . The same pairing carries the Hawking sector information in Doc. 325.

**(F3) Finite fractal recursion.** The fractal measure arises from a finite 100-step recursion (Doc. 322), hence  $0 < c \leq w \leq C$  for the weight  $w$  — bounded and strictly positive.

## .3 Proof chain

### Lemma 1 — Filtration realises the Doc.-322 axioms [B]

Under (F1), the  $L_n$  satisfy exactly  $L_n^2 = L_n$ ,  $L_n^\dagger = L_n$ ,  $L_n L_m = L_{\min(n,m)}$ : the “fractal inclusion structure” of Doc. 322 is a scale filtration. (*Script [1]: exact to machine precision.*)

### Theorem 2 — Diagonal form and boundedness [B]

Under (F1)+(F2),  $\hat{F}$  is block-diagonal:

$$\hat{F}|_{\mathcal{H}_k} = \varphi_k \cdot \mathbb{1}, \quad \varphi_k = \sum_{n \geq k}^N r_n e^{i\theta_{n,k}}.$$

By the finite rung count (F1):  $\|\hat{F}\| \leq \sum_{n=1}^N r_n < \infty$ .  $\hat{F}$  is **bounded**,  $\mathcal{D}(\hat{F}) = \mathcal{H}$ . (Script [2]+[3].)

### Theorem 3 — $\mathbb{Z}_3$ pairing generates symmetry [B]

Under (F2), every phase appears in the pair  $(e^{i\theta}, e^{-i\theta})$ . The orbifold-covariant operator has real sector values:

$$\varphi_k^{\text{cov}} = \sum_{n \geq k}^N r_n \cos \theta_{n,k} \in \mathbb{R},$$

hence  $\hat{F} = \hat{F}^\dagger$ . **Symmetry is a  $\mathbb{Z}_3$  consequence, not a requirement.** Counter-example: one unpaired phase gives  $\|\hat{F} - \hat{F}^\dagger\| \approx 0.02 \neq 0$ . (Script [4].)

### Corollary 4 — Self-adjointness, deficiency indices $(0, 0)$ [B]

Bounded (Thm. 2) + symmetric (Thm. 3)  $\Rightarrow$  self-adjoint on  $\mathcal{H}$ . Deficiency indices:  $\ker(\hat{F}^\dagger \mp i) = \{0\}$ , since  $\|(\hat{F} \mp i)^{-1}\| \leq 1$  for bounded self-adjoint  $\hat{F}$ . **Exactly one** self-adjoint realisation — no extension choice. (Script [5]: *smallest singular value exactly 1*.)

### Theorem 5 — Tensor factor $\hat{\Delta}_{T^4}$ with fractal measure [B]

1.  $\Delta$  on compact  $T^4$  (no boundary): essentially self-adjoint, Fourier eigenbasis with eigenvalues  $|n|^2 \geq 0$ . (Standard.)
2. Fractal measure (F3):  $0 < c \leq w \leq C$ , so  $\psi \mapsto \sqrt{w}\psi$  is a unitary isomorphism  $L^2(d\mu_f) \rightarrow L^2(d\mu)$ . The fractal measure *preserves* self-adjointness. (Script [6].)

### Theorem 6 — $\mathbb{Z}_3$ restriction and $\chi$ -twists [B]

1.  $[\hat{F}, P_0] = 0$  ( $\mathbb{Z}_3$  covariance from (F2))  $\Rightarrow$  restriction to the orbifold sector is self-adjoint. (Script [7].)
2. Fixed-point boundary conditions  $\chi^3 = 1$  (Docs. 322/320): unitarily implemented twists; spectrum  $(n + \alpha)^2 \geq 0$  real in all three twist classes — each carries a self-adjoint realisation. (Script [8].)

### Corollary 7 — $\xi$ as well-defined minimum [B]

$\hat{F}_{D_4}$  (restriction to the  $D_4$  sector, Doc. 314) is self-adjoint. By the min-max principle:

$$\xi = \lambda_{\min}(\hat{F}_{D_4}) = \min_{\|\psi\|=1, \psi \in \mathcal{H}_{D_4}} \langle \psi, \hat{F}\psi \rangle$$

**well-defined and truncation-stable:** the lowest mode lies in every truncation space that contains it. (Script [9].)

## .4 Status table

| Statement                                                             | Status      |
|-----------------------------------------------------------------------|-------------|
| $L_n L_m = L_{\min}$ realised by filtration, exactly                  | <b>[B]</b>  |
| $\hat{F}$ bounded by finite rung count ( $L_0$ )                      | <b>[B]</b>  |
| Symmetry as $\mathbb{Z}_3$ consequence, with counter-example          | <b>[B]</b>  |
| Self-adjointness, deficiency indices $(0, 0)$ , uniqueness            | <b>[B]</b>  |
| Fractal measure (finite recursion) preserves self-adjointness         | <b>[B]</b>  |
| $\mathbb{Z}_3$ restriction and all three $\chi$ -twist classes s.a.   | <b>[B]</b>  |
| $\xi = \lambda_{\min}(\hat{F}_{D_4})$ well-defined, truncation-stable | <b>[B]</b>  |
| Remaining cases                                                       | <b>none</b> |

The R82 gap is fully closed. There are no infinities in FFGFT;  $\sum r_n < \infty$  was never an extra requirement — it follows directly from  $L_0 = \xi \cdot \ell_p$ .

## .5 Proposed register entry R86

**R86 – Self-adjointness of  $\hat{F}$  (August 2026):** Doc. 327 closes the gap declared as [S] in R82 completely. The finiteness of the scale ladder is guaranteed by  $L_0 = \xi \cdot \ell_p$  (Doc. 180 **[K]**) – no infinities in FFGFT, hence  $\hat{F}$  bounded and  $\mathcal{D}(\hat{F}) = \mathcal{H}$ . The symmetry  $\hat{F} = \hat{F}^\dagger$  follows from the  $\mathbb{Z}_3$  pairing  $(k, -k)$  (the same as in Doc. 325 **[B]**). Deficiency indices  $(0, 0)$ : exactly one self-adjoint realisation **[B]**. Fractal measure (finite 100-step recursion),  $\mathbb{Z}_3$  restriction and all three  $\chi$ -twist classes preserve self-adjointness **[B]**.  $\xi = \lambda_{\min}(\hat{F}_{D_4})$  well-defined and truncation-stable **[B]**. No remaining cases. Verification script: 327\_selbstadjungiertheit\_F\_verify.py (21 assertions, all passed).

*Declared 23 August 2026*

## Relation to the corpus

| Building block                                             | Doc.     |
|------------------------------------------------------------|----------|
| Minimal length scale $L_0 = \xi \cdot \ell_p$              | 180      |
| Definition of $\hat{F}$ , $\mathcal{D}(\hat{F})$ , R82 gap | 322      |
| $D_4$ lattice in Hilbert space                             | 314      |
| Fixed-point boundary conditions, neutrino localisation     | 320, 322 |
| $\mathbb{Z}_3$ triality, sector structure                  | 321      |
| Sector pairing $(k, -k)$ — same as in Hawking mechanism    | 325      |
| Fractal measure, 100-step recursion, $D_f = 3 - \xi$       | 322      |

## References

1. J. Pascher, Docs. 180, 314, 320, 321, 322, 325, FFGFT corpus, 2026.

2. M. Reed, B. Simon, *Methods of Modern Mathematical Physics I/II*, Academic Press.
3. J. von Neumann, *Allgemeine Eigenwerttheorie Hermitescher Funktionaloperatoren*, Math. Ann. 102, 1929.

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